KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2019
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Why is the pattern matching by regular expression greedy? Explain with example. (09)
   b) Define text normalization. What is the difference between lemmatization and stemming? Explain with example. (10)
   c) Explain porter's algorithm with example. (11)
   d) What is types and tokens? How many types and tokens are there in the following sentence? "He stepped out into the hall, was delighted to encounter a water brother." (05)

2. a) "Finding the minimum edit distance is problem of the category 'dynamic programming' "- Explain the statement to convert one String X of length n to a String Y of length m. (12)
   b) Apply the minimum edit distance algorithm (using insertion cost = 1, deletion cost = 1 and substitution cost = 2) of "leda" to "deal". Show your work in edit distance grid. (10)
   c) Define bag of words problem. How is bag of words conditional independence assumption applied in Naïve Bayes text classification? Explain. (13)

3. a) Define Stop word. How do you handle Stop word in text classification? (08)
   b) Given the following text classification for classes C_1 and C_2 (10)
      i) PQR $\rightarrow C_1$
      ii) PPS $\rightarrow C_1$
      iii) PT $\rightarrow C_1$
      iv) UVP $\rightarrow C_2$

      Compute the most likely class for the text "PPPUV".
   c) Consider the following likelihood for a review (05)

	Pos	Neg
I	0.09	0.16
always	0.07	0.06
like	0.29	0.06
foreign	0.04	0.15
films	0.08	0.11

   What class will the Naïve Bayes assign to the sentence "I always like foreign films"?
   d) Find the context free rules and hence the context free grammar (CFG) for the following English sentences. (12)
      i) I want a morning flight.
      ii) Which flight serves breakfast?
      iii) Show the lowest fare.
      iv) The flight should be at 11 am.

4. a) Consider the following grammar in CNF (10)

$$S \rightarrow AB \mid BB$$
$$A \rightarrow CC \mid AB \mid a$$
$$B \rightarrow BB \mid CA \mid b$$
$$C \rightarrow BA \mid AA \mid b$$

   Is 'aabb' in L(G)? Justify your answer using CYK algorithm.
   b) How does Early Parser differ from CYK Parser? Explain finite state chunking with example. (10)
   c) Define PCFG? How can you disambiguate a syntactic Parsing by PCFG? (06)
   d) Explain the IR based factoid question answering using example. (09)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define Natural Language Processing (NLP). Why do we need NLP? Write some industrial applications of NLP? Derive (10)
   b) What does smoothing mean? ~~Drive~~ the equation for Laplace smoothing. (10)
   c) Given the following corpus: (07)

   I like NLP.
   I like Deep Learning.
   NLP and Deep Learning have huge area for research.

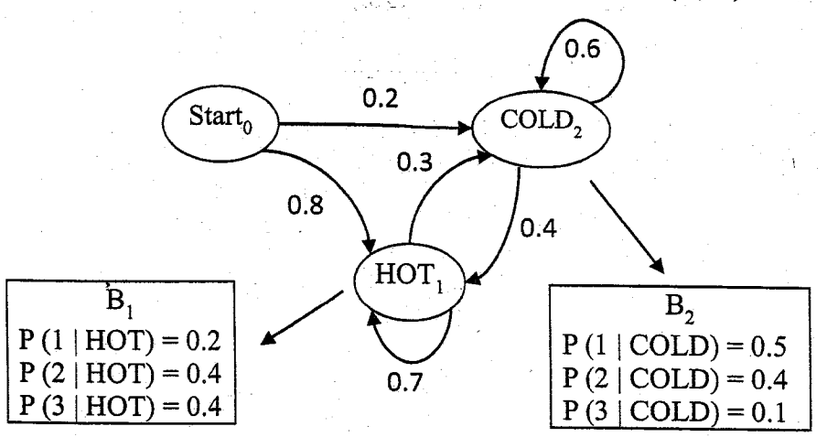
   Using Bigram Language model with add-one smoothing, what is P (NLP | like)?
   d) Define zero situation and discuss the effect of it. (08)

6. a) "POS tagging is a disambiguation task" – Justify the statement with example. (08)
   b) Define perplexity. Derive the equation of computing W with a Bigram Language model. (10)
   c) For Hidden Markov Model (HMM) Pos tagging, using the following formula, find the equation of calculating tag transition probabilities (10)

$$\hat{t}_1^n = \underset{t_1^n}{argmax}\, P\,(t_1^n \mid w_1^n)$$

   d) Consider the sentence: "Eight/CD horses/NNS will/MD race/? For/IN the/DT cup./NN." Given the probabilities below, find the right Pos tag for the word "race".(VB or NN). (07)
   P (VB | MD) = 0.0045, P (NN | MD) = 0.062, P (race | NN) = 0.048, P (race | VB) = 0.014, P (IN | VB) = 0.0012, P (IN | NN) = 0.0024

7. a) Define Markov chain. Explain the components of Markov chain. (06)
   b) For logistic regression, show that P (y = true | x) = 1 | (1 + $e^{-w.f}$) (10)
   c) Given a sequence of ice-cream observations 3 1 3 and an HMM λ = (A, B) in the following figure. (12)



   Find the best hidden weather sequence Q (like H H H).
   d) What are the statistical models for processing text and speech? Give example of fully connected & left to right HMM. (07)

8. a) Define sentence tokenization. Draw the architecture of a TTS. (07)
   b) Discuss about the steps of speech synthesis (text to waveform). Show the Hourglass Metaphor. (10)
   c) What is prosody? Discuss about the three aspects of prosody. (10)
   d) What is accent ratio? Why is accent ratio important? (08)